

From Cointegration to Out-of-Sample Failure: A Pairs-Trading Case Study on PEP/KO

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Abstract

This report examines whether a cointegration-based pairs trading strategy between PepsiCo and The Coca-Cola Company is statistically robust and economically exploitable. Using daily data from 2013 to 2023, we first test for cointegration and estimate the spread's mean-reversion dynamics over 2013–2018, then hold these statistical parameters fixed and optimise a threshold-based trading strategy in-sample over 2018–2023. Robustness is assessed through transaction-cost and parameter sensitivity tests, walk-forward validation, and Adjusted and Deflated Sharpe Ratios. The strategy is then evaluated out-of-sample from 2023 to the present, including an analysis of time-varying hedge ratios using rolling OLS and a Kalman filter. The results show that weakening mean-reversion dynamics in the spread undermine the effectiveness of the strategy out-of-sample.

1 Introduction

Pairs trading is a market-neutral strategy that seeks to profit from temporary mispricings between two historically related assets. A common statistical foundation for the strategy is *cointegration*: although two asset prices may each be non-stationary, a linear combination of them can be stationary. This long-run trend suggests that deviations from their equilibrium relationship may eventually reverse. If the relationship is sufficiently stable, such deviations can potentially be exploited through a systematic trading strategy.

In this report we investigate this hypothesis empirically for a canonical pair of consumer-staples competitors, PepsiCo (PEP) and The Coca-Cola Company (KO), using daily price data from 2013 to the present. The central question we address is whether a statistically significant cointegrating relationship is stable and economically exploitable once transaction costs, model-selection bias, and regime

change are taken into account.

The analysis proceeds in four stages. The first step is classical: we examine the stationarity properties of the individual log-price and return series using the 2013–2018 daily price data using the Augmented Dickey–Fuller test, and then apply the Engle–Granger procedure to test for cointegration. The resulting spread is further characterised as an $AR(1)$ model, from which we estimate its mean-reversion speed and half-life. Second, we exploit these findings to justify a rule-based trading policy governed by entry and exit thresholds on the standardised spread, and we optimise this policy in-sample over a 2018–2023 training window. Third, we examine the robustness of the resulting strategy through sensitivity tests on transaction costs, risk budgets, and volatility-window lengths, as well as through walk-forward validation. We also investigate the effect of the COVID-19 volatility shock and apply both the Adjusted Sharpe Ratio and Deflated Sharpe Ratio to account for non-Gaussian returns and the selection bias associated with searching across multiple candidate policies. Finally, the selected strategy is evaluated out-of-sample (OOS) from 2023 to the present. We assess whether the hedge ratio estimated during the 2013–2018 period remains appropriate over time and consider two approaches that allow it to evolve dynamically: rolling OLS and a Kalman filter.

Throughout, particular care is taken to avoid look-ahead bias and to distinguish between statistical parameters, estimated using data from the 2013–2018 window, and trading parameters, which are optimised in-sample during the 2018–2023 training window and subsequently evaluated (but not re-optimised) OOS. Moreover, performance statistics such as the Sharpe ratio are carefully interpreted in light of the number of trades and the number of policies searched over.

Remark 1. Throughout, prices are retrieved as adjusted daily closing prices, i.e. corrected for dividends and stock splits. Unadjusted prices exhibit a drop on each ex-dividend date that is unrelated to the underlying price dynamics which we seek to model. Left uncorrected, such drops would appear as spurious negative jumps in the return series r_t , potentially triggering false trading signals in Section 3.

2 Stationarity and Cointegration

The statistical analysis is based on the prices of PEP and KO over the period 2013–2018. Let P_t denote the closing price of a given company at time t , and denote its log-price and log-return by $X_t := \log P_t$ and $r_t := X_t - X_{t-1}$, respectively. Over the sample period, the annualised volatility of PEP and KO log-returns is 13.09% and 13.80% respectively, consistent with two large, mature and low-growth companies. Their daily log-returns exhibit a sample correlation of $\hat{\rho}_{\text{PEP},\text{KO}} = 0.66$. Although the two return series exhibit a strong contemporaneous association, correlation alone does not establish a long-run relationship between the underlying price processes. Since correlations between non-stationary series may be spurious, determining the order of integration of each price series is a necessary first step before testing for cointegration.

We model the log-price process as a first-order autoregressive process, $AR(1)$,

$$X_t = \alpha + \rho X_{t-1} + \varepsilon_t, \quad \varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2),$$

where we assume $\text{Cov}(X_0, \varepsilon_t) = 0$ for all t . The qualitative behaviour of the process is governed by ρ : the process is weakly stationary if $|\rho| < 1$, has a unit root if $\rho = 1$, and is explosive if $|\rho| > 1$.

2.1 Unit Root Testing

Before conducting a formal unit-root test, a qualitative inspection of the price dynamics rules out, to a reasonable approximation, both explosive and alternating behaviour. We therefore restrict attention to the case $\rho \in (-1, 1]$. Within this restricted parameter space, testing $H_0 : \rho = 1$ against $H_1 : \rho < 1$ constitutes a binary hypothesis test, which we carry out using the Augmented Dickey–Fuller (ADF) test. The lag order is selected according to the Akaike Information Criterion (AIC), which trades off goodness of fit against model complexity. Applied to the log-price series, the ADF test fails to reject the unit-root null for either company at the 5% level. Con-

versely, applying the same test to the log-return series $r_t = \Delta X_t$ results in a strong rejection of the unit-root null for both companies at the 5% level (see Table 1 for both tests). Taken together, these results indicate that both log-price series are $I(1)$: they are non-stationary in levels but stationary after first differencing.

Series	ADF statistic	p-value	5% CV
$\{X_t^{\text{PEP}}\}$	−1.545	0.511	−2.864
$\{X_t^{\text{KO}}\}$	−1.663	0.450	−2.864
$\{r_t^{\text{PEP}}\}$	−36.50	< 0.001	−2.864
$\{r_t^{\text{KO}}\}$	−36.62	< 0.001	−2.864

Table 1: ADF test results for log-prices and log-returns, 2013–2018.

2.2 Cointegration and Spread Dynamics

Since both log-price series are $I(1)$, we next investigate whether they are cointegrated in the sense of Engle and Granger. In particular, cointegration asks if there exist $\alpha, \beta \in \mathbb{R}$ such that the linear combination

$$u_t := X_t^{\text{PEP}} - \beta X_t^{\text{KO}} - \alpha$$

is stationary. We estimate (α, β) by OLS regression of X_t^{PEP} on X_t^{KO} , obtaining $\hat{\alpha} = -1.409$ and $\hat{\beta} = 1.662$. We then apply the Engle–Granger test to the estimated residuals $\hat{u}_t = X_t^{\text{PEP}} - \hat{\beta} X_t^{\text{KO}} - \hat{\alpha}$ – we use the adjusted critical values to account for the fact that the cointegrating relationship is estimated from the data. The adjusted test rejects the null hypothesis of no cointegration at the 5% significance level (p -value 0.033), providing evidence of a long-run equilibrium relationship between the PEP and KO log-prices over the sample period.

Having established the stationarity of $\hat{u}_t$, we characterise its dynamics by fitting the $AR(1)$ model

$$\hat{u}_t = a + \rho_u \hat{u}_{t-1} + \eta_t,$$

which yields $\hat{\rho}_u = 0.9785$ and an implied equilibrium level

$$\hat{m} = \frac{\hat{a}}{1 - \hat{\rho}_u} = 0.0037,$$

which is effectively zero. The corresponding mean-reversion speed $\hat{\kappa} := 1 - \hat{\rho}_u = 0.0215$ implies a half-life of mean reversion of approximately 31.9 trading days. Together, the estimated cointegrating relationship and the dynamics of the stationary spread

provide the statistical basis for the entry, exit, and volatility-window parameters of the trading policy developed in the following section.

3 Trading Policy

We translate the estimated spread dynamics into a rule-based trading policy comprising entry and exit thresholds, position sizing, and an explicit treatment of transaction costs.

Entry and exit. We standardise the estimated spread $\hat{u}_t$ into a z -score,

$$z_t := \frac{\hat{u}_t - \hat{m}}{\hat{\sigma}_u},$$

where $\hat{m}$ and $\hat{\sigma}_u$ are, respectively, the equilibrium level and sample standard deviation of the spread estimated previously. Hence, z_t measures how many standard deviations the spread currently lies from its long-run equilibrium. A large positive z_t indicates an unusually high spread: under the mean-reversion assumption, it is expected to decline toward equilibrium; a large negative z_t indicates an unusually low spread and is expected to rise. The trading policy is therefore governed by $\theta := (z_{\text{entry}}, z_{\text{exit}})$, with $z_{\text{entry}} > 0$ and $0 \leq z_{\text{exit}} < z_{\text{entry}}$. When z_t crosses above z_{entry} , the strategy enters a *short-spread* position (short one unit of PEP, long $\hat{\beta}$ units of KO), which profits from a subsequent decline in the spread; when z_t crosses below $-z_{\text{entry}}$, it enters the opposite *long-spread* position. In both cases, the position is closed once $|z_t|$ falls below z_{exit} , at which point the spread has reverted sufficiently close to equilibrium that any further reversion no longer justifies holding the position.

Position sizing. Sizing is fixed ex ante by a risk-budgeting rule and is not an optimisation parameter. Let C_t denote total portfolio equity at time t , inclusive of both realised and unrealised profits and losses, with initial capital $C_0 = \$100,000$, and let w denote the fixed fraction of equity allocated as nominal risk budget at each time step, $w \cdot C_t$. The position is scaled inversely to the recent volatility $\hat{\sigma}_{\Delta z}^L$ of the spread's daily changes $\Delta z_t = z_t - z_{t-1}$, estimated over a trailing window of L days up to $t-1$ to avoid look-ahead. Because z_t is standardised, $\hat{\sigma}_{\Delta z}^L$ does not capture the magnitude of the spread's movements. The additional factor $\hat{\sigma}_u$ is therefore included to restore the scale of the underlying spread's volatility.

The resulting dollar notional exposure is

$$N_t = C_t \min \left(\frac{w}{\hat{\sigma}_u \hat{\sigma}_{\Delta z}^L}, N_{\max} \right),$$

where $N_{\max}$ caps the maximum fraction of current equity for the position. For a long position in the spread, the corresponding share counts are therefore $S_t^{\text{PEP}} = N_t / P_t^{\text{PEP}}$ and $S_t^{\text{KO}} = -\hat{\beta} N_t / P_t^{\text{KO}}$, with signs reversed for a short position.

Transaction costs. A transaction cost c , expressed as a fraction of traded notional, is applied once on entry and once on exit of each leg. The costs are deducted directly from C_t , so that they enter the objective function (see below) directly rather than being a post-hoc adjustment. This is especially important because narrower entry/exit bands capture smaller reversions but trigger more frequent round-trip trades, so a cost-blind criterion can favour more active policies.

Optimisation objective. Let

$$\Pi_t(\theta) = \frac{C_t(\theta) - C_{t-1}(\theta)}{C_{t-1}(\theta)}$$

denote the daily return on capital for a given policy θ . The optimal policy θ^* is selected by maximising the annualised net Sharpe ratio

$$\theta^* \in \arg \max_{\theta \in \Theta} J(\theta), \quad J(\theta) = \frac{\hat{\mu}_{\Pi}(\theta)}{\hat{\sigma}_{\Pi}(\theta)} \sqrt{252}, \quad (1)$$

over a predefined grid of candidate policies Θ . Raw net P&L is deliberately avoided as an objective, since it may reward highly levered or infrequent profitable trades without penalising risk.

Equity Curve $C_t(\theta^*)$ under Optimal Trading

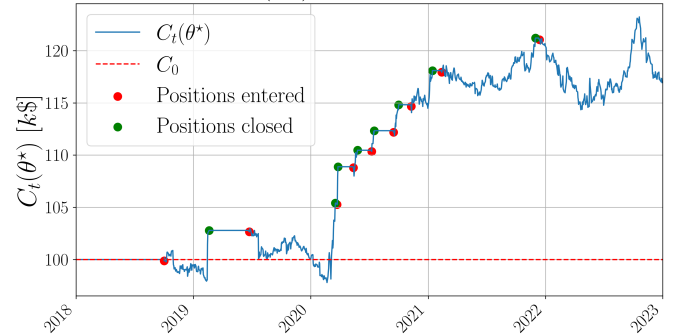


Figure 1: Equity curve under optimal trading during the 2018–2023 training window.

3.1 Policy Optimisation Results

The policy parameters other than θ are fixed ex ante, with a risk budget $w = 0.5\%$, a maximum notional exposure $N_{\max} = 25\%$, a transaction cost $c = 0.20\%$, and a volatility-estimation window $L = 64$ trading days (approximately twice the mean-reversion half-life estimated in Section 2). Optimising J over the threshold grid, using only the 2018–2023 training window, selects

$$\theta^* = (z_{\text{entry}}^*, z_{\text{exit}}^*) = (2.15, 0.50),$$

generating 9 round-trip trades, an annualised net Sharpe ratio of $J(\theta^*) = 0.67$, and a net P&L of \$16,941.20 on the initial capital of \$100,000. Figure 1 depicts the equity curve under optimal trading.

4 Robustness Analysis

We first assess the robustness of the optimal policy $\theta^* = (2.15, 0.50)$ selected in Section 3 to its auxiliary parameters. We then evaluate the stability of the optimisation procedure itself using walk-forward testing.

4.1 Sensitivity Analysis

We vary each auxiliary parameter (transaction cost c , risk budget w , volatility-estimation window L , and maximum exposure $N_{\max}$) individually while holding the remaining parameters at their baseline

values: $c = 0.20\%$, $w = 0.50\%$, $L = 64$ days, $N_{\max} = 25\%$. For each parameter set, the policy θ is re-optimised as in (1).

	Range	$\theta^* = (2.15, 0.50)$ for	J^* (range)
c	0.01–1.00%	$0.05\% \leq c \leq 0.50\%$	0.32–0.77
w	0.10–1.00%	$w \geq 0.50\%$	0.55–0.72
L	3–478 days	$L \geq 6$ days	0.67–0.72
$N_{\max}$	5–100%	$N_{\max} \leq 25\%$	0.55–0.72

Table 2: Sensitivity of θ^* to auxiliary parameters.

The baseline policy $(2.15, 0.50)$ remains optimal across the majority of the parameter grid (see Table 2). At the boundaries of some grids, the optimum shifts toward lower entry and exit thresholds, corresponding to more frequent trading. This occurs at low transaction costs and risk budgets, shorter volatility windows, and higher exposure limits, where the penalty associated with increased trading activity or position size is reduced. The resulting variation in J^* is moderate across most parameter ranges. Overall, the recurrence of the same θ^* across broad ranges of the auxiliary parameters provides evidence that the selected policy is not highly sensitive to their precise calibration.

Walk-Forward Validation: Equity Curve $C_t(\theta_k^*)$ for each Fold k

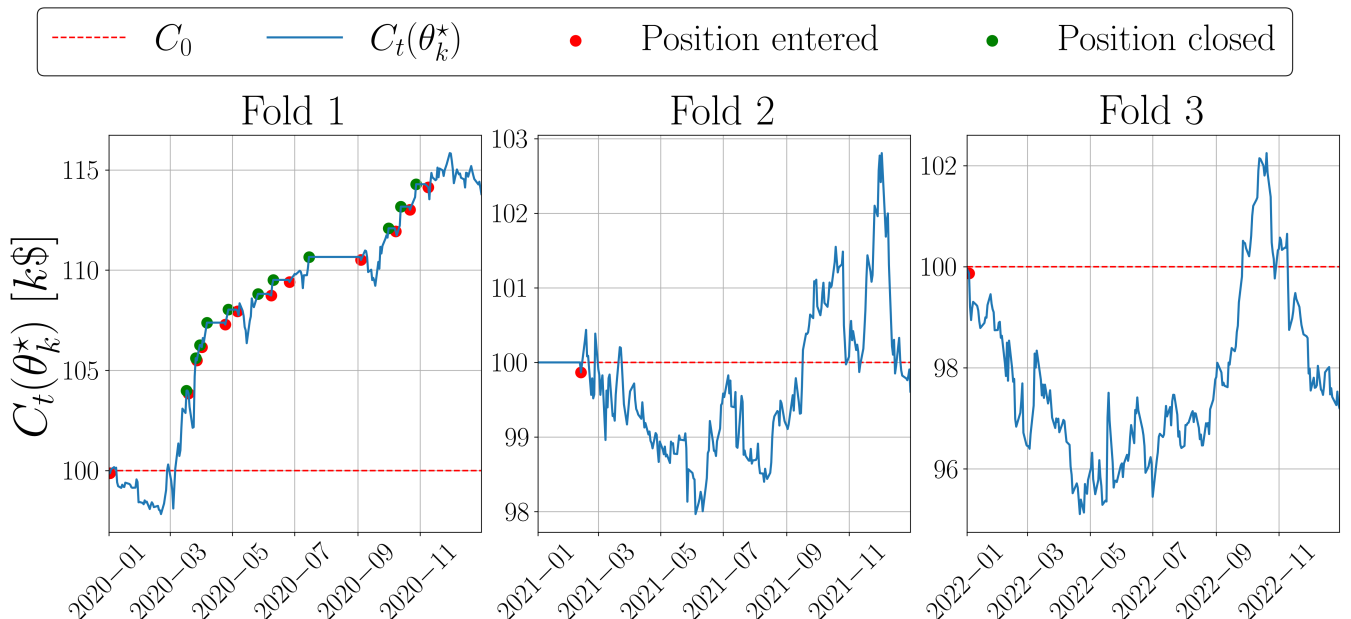


Figure 2: Walk-forward validation equity curve during the OOS window $\mathcal{S}_k$ for each fold $k = 1, 2, 3$.

4.2 Walk-Forward Validation

To assess whether the optimised policy generalises beyond the 2018-2023 sample used for calibration, we conduct a walk-forward analysis. For each of three folds, we re-optimize θ using a rolling two-year training window $\mathcal{T}_k$ and then apply the resulting policy, without further adjustment, to the following one-year OOS window $\mathcal{S}_k$. The folds are defined as $\mathcal{T}_1 = [2018, 2020)$ through $\mathcal{T}_3 = [2020, 2022)$, with each training window followed by its corresponding one-year test window. We then concatenate the three OOS return series and compute the aggregate OOS Sharpe ratio J^{OOS} .

Fold k	z_{entry}^*	z_{exit}^*	$J_{\mathcal{S}_k}^*$	Net P&L	n_{trades}
1	1.30	0.65	2.14	\$13,793.67	12
2	2.15	0.35	-0.07	-\$391.21	1
3	2.00	0.50	-0.50	-\$2,795.69	1

Table 3: Walk-forward OOS results by fold.

The OOS results are highly uneven across folds (see Figure 2). Fold 1 generates most of the aggregate performance, with 12 trades and a Sharpe ratio of 2.14, whereas folds 2 and 3 each generate only one trade and negative returns. The optimal thresholds also vary across folds, indicating that the calibrated policy is sensitive to the training period. Overall, the aggregate OOS Sharpe ratio $J^{\text{OOS}} = 0.64$ provides evidence of positive performance outside the calibration sample, but the strong variation across folds suggests that this performance is not consistent across time.

4.3 COVID-19 Dislocation and Regime-Dependence

The pronounced asymmetry across walk-forward folds is closely associated with the COVID-19 volatility dislocation. An inspection of the optimal equity curve over the full 2018–2023 sample shows that a substantial share of total net P&L is generated during the February–April 2020 period, characterised by an unusually large and rapid increase in volatility followed by sharp mean reversions – exactly the kind the strategy is designed to exploit. The concentration of performance in this period therefore raises concerns about regime dependence and the extent to which the full-sample results generalise to more typical market conditions.

To isolate this, we re-run the analysis with

COVID-19 window exclusion:
aggregate annualised net Sharpe ratio

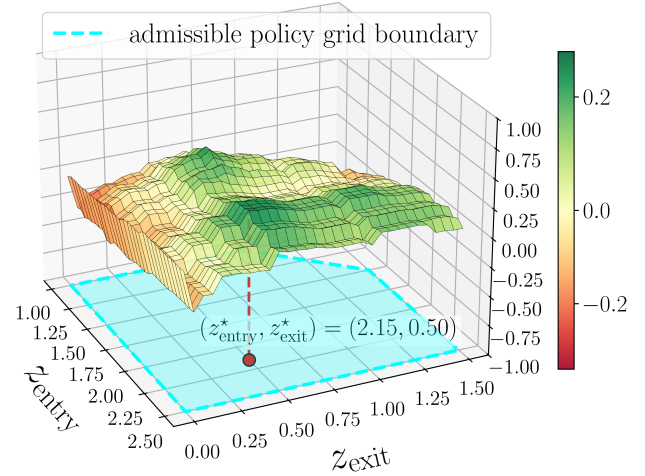


Figure 3: Aggregate annualised net Sharpe ratio surface over the admissible grid Θ during the Pre- and Post-COVID windows.

the COVID window excluded, treating the pre- and post-COVID periods as two independent segments, each with its own rolling-volatility buffer and equity reset. This prevents the exclusion window from creating an artificial jump in returns or contaminating the rolling volatility estimates at the splice point.

The optimiser still selects $\theta^* = (2.15, 0.50)$, but performance deteriorates materially; see Figure 3 for the aggregate annualised net Sharpe ratio over the admissible grid. Under optimal trading, the trade count falls from 9 to 8, the annualised Sharpe ratio from 0.67 to 0.28, and net P&L from \$16,941 to \$5,759. With only 8 trades, these results carry considerable sampling uncertainty, but the direction of the effect is unambiguous: a large share of the strategy’s headline performance is attributable to the COVID dislocation, and its edge outside acute volatility regimes appears modest at best.

4.4 Adjusted and Deflated Sharpe Ratios

The Sharpe ratios reported so far have two limitations. First, they assume Gaussian returns, whereas the daily return on capital is mostly zero with occasional large moves. Second, θ^* was selected as the best-performing policy among $N = 901$ candidates. We therefore compute the Adjusted Sharpe Ratio (ASR), which accounts for skewness and excess kurtosis using the Pezier–White adjustment, and the Deflated Sharpe Ratio (DSR), which accounts for the multiple-policy search by comparing the observed

daily Sharpe ratio with the level expected from N zero-skill strategies. Both measures are computed on the COVID-excluded sample.

The ASR is 0.281, essentially unchanged from the raw Sharpe ratio of 0.28, indicating that non-normality has little effect on the risk-adjusted performance estimate at this Sharpe level. The DSR is 0.423, implying that the observed Sharpe ratio is not statistically distinguishable from the maximum Sharpe ratio that could arise by chance when searching over 901 zero-skill policies. While this does not establish that the strategy has no genuine edge, it indicates that the in-sample optimisation provides limited statistical evidence of one, consistent with the weak performance observed after excluding the COVID-19 dislocation.

5 Out-of-Sample Performance

We evaluate $\theta^* = (2.15, 0.50)$ on the 2023–present period as an OOS test. The statistical parameters $\hat{\beta}, \hat{\alpha}, \hat{m}, \hat{\sigma}_u$ are frozen at the estimates obtained from the statistical analysis (2013–2018), while the policy parameters $c, w, N_{\max}, L$ are frozen at their baseline values (see Section 3). At the trade frequency observed in-sample during the 2018–2023 window (roughly 2–3 trades per year outside the COVID window), this window is expected to yield only 5–10 trades, too few for any resulting Sharpe ratio to be treated as a reliable performance estimate.

The results of the OOS test are reorted in Table 4. The strategy enters a single position early in the window and force-liquidates it at the end, as the exit threshold z_{exit} is never crossed. Equity declines almost monotonically from a few weeks after entry, resulting in a net P&L of approximately $-\$27,000$ before liquidation. With only one trade, the reported Sharpe ratio of -1.16 is not a meaningful estimate of OOS performance. However, the underlying failure mode is informative. Re-estimating the hedge ratio on 2023–present data yields $\hat{\beta}_{\text{OOS}} = -0.2376$, compared with $\hat{\beta} = 1.6618$ in the 2013–2018 period. Thus, the estimated PEP/KO relationship changes not only in magnitude but also in sign. A post-hoc Engle–Granger test on the resulting residuals remains marginally significant ($p = 0.039$). This evidence suggests hedge-ratio instability contributing to the deterioration in OOS performance, although the single-trade sample does not allow a definitive attribution.

J^*	Net P&L	n_{trades}
-1.16	$-\$27,371.58$	1

Table 4: Out-of-sample test (2023–present) with frozen parameters.

5.1 Rolling Hedge Ratio

The sign flip in $\hat{\beta}_{\text{OOS}}$ motivates a diagnostic test using a time-varying hedge ratio β_t . We consider two standard estimators: *rolling OLS*, which re-estimates (α_t, β_t) over a trailing M -day window ($M = 504$, approximately two trading years); a *Kalman filter*, which models β_t as a random walk,

$$\beta_t = \beta_{t-1} + w_t, \quad w_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_w^2),$$

and recursively updates $\hat{\beta}_{t|t}$ from the observation equation

$$X_t^{\text{PEP}} = \alpha + \beta_t X_t^{\text{KO}} + \varepsilon_t.$$

Both estimators use only information available up to $t - 1$, preserving the no-look-ahead property of the static backtest. However, neither is subjected to the same walk-forward, sensitivity, or ASR/DSR analysis as the static model. The results are therefore interpreted as diagnostic rather than as evidence of a validated alternative strategy.

Estimator	J^*	Net P&L	n_{trades}
Rolling OLS ($M = 504$)	-1.06	$-\$918.41$	1
Kalman filter	-0.84	$-\$251.17$	5

Table 5: OOS test (2023–present) with adaptive hedge ratio and static policy $\theta^* = (2.15, 0.50)$.

Both estimators indicate a similar evolution in the hedge ratio. As shown in Figure 4, rolling OLS estimates a decline in β_t from approximately 1.0 to -0.5 before increasing again, consistent with the sign change observed in the full-period OOS estimate. The Kalman filter also captures the hedge ratio drift, though with a significantly smoother trajectory than rolling OLS. Tested OOS over the 2023–current window with the frozen policy $\theta^* = (2.15, 0.50)$ and the policy parameters at their baseline values, both estimators substantially reduce losses relative to the backtest with a static hedge ratio, which is consistent with an adaptive β_t tracking the evolving relationship between the two assets more closely at entry.

With only $n = 1$ and $n = 5$ trades, however, neither Sharpe ratio is statistically meaningful.

Neither estimator renders the strategy profitable. The results presented above provide evidence (although weak) that hedge ratio drift is a major contributor to the failure of the static model over the 2023–present window, since both hedge ratio estimators reduce the loss by more than an order of magnitude. They do not, however, establish drift as the sole explanation, since both equity curves continue to decline after mid-2024 while a position is open. This suggests a residual source of failure, for instance a spread that no longer exhibits the mean reversion assumed by the policy.

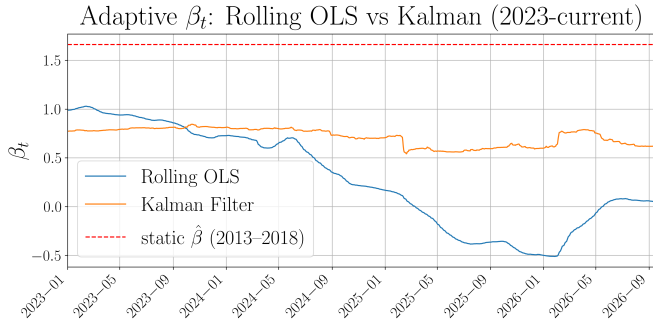


Figure 4: Time-varying hedge ratio β_t , 2023–present: rolling OLS vs. Kalman filter.

6 Conclusion

This report examined whether a cointegration-based PEP/KO pairs trading strategy remains statistically robust and economically exploitable once transaction costs, model uncertainty, and policy selection are taken into account. The evidence presented in this report provides limited support for this proposition.

The 2013–2018 analysis in Section 2 identifies a statistically significant cointegrating relationship between PEP and KO, with a slowly mean-reverting spread (half-life ≈ 32 days), providing a basis for the policy developed over the 2018–2023 optimisation period in Section 3. Section 4 shows that, although the resulting policy is relatively robust to its auxiliary parameters, walk-forward validation exposes that its performance is concentrated in a single fold; excluding the COVID-19 market dislocation reduces the Sharpe ratio by more than half. After accounting for the 901 policies considered, a Deflated Sharpe Ratio of 0.527 further indicates that the in-sample results provide limited evidence of genuine skill.

The 2023–present out-of-sample test in Section 5 provides further evidence against the robustness of the strategy. The strategy generates a single losing trade and a substantial loss, while the hedge ratio changes markedly and reverses its sign relative to its 2013–2018 estimate. Rolling OLS and Kalman filter diagnostics show a similar evolution in the hedge ratio. Allowing the hedge ratio to vary over time reduces the realised loss in this OOS period, but does not eliminate it. Given the single trade, however, these results should not be interpreted as evidence for a viable adaptive strategy.

More broadly, the results illustrate that a statistically significant and economically plausible cointegrating relationship within one estimation regime does not imply that its trading dynamics will remain stable out of sample. For PEP and KO, the mean-reversion characteristics observed during the estimation and optimisation periods did not persist into 2023–present, and the concentration of in-sample performance around the COVID-19 volatility regime further limits the evidence for a durable trading opportunity. Real-time regime-adaptive hedging is a natural direction for further investigation, but the pair’s low trading frequency would continue to limit statistical power and make reliable validation challenging.